THE CHERN CLASSES, THE CHERN CHARACTER COMPONENTS AND THE POWER SUMS OF A PERFECT OBJECT ARE MULTIPLIED BY THE SIGN $(-1)^i$ WHEN ONE PASSES TO THE DUAL

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file chow.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with \input{preamble} and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-chow/preamble.tex is excerpted verbatim as three spans, lines 8-12 (the verbatim package together with the project's own reference, slogan and history display-environment definitions, including the author's commented-out older form of the reference environment), lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). The lines 8-12 span is required because a counted statement or its surrounding text may use those display environments, which the project typesets as comments. Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. Same-chapter pointers are located at their \label lines inside the pinned chapter file; cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Allowed standard prerequisites (editorial).

- The reduction of statements about a perfect object to a bounded complex of finite locally free modules of fixed rank. Same chapter chow.tex, label lemma-commutative-chern-perfect (invoked at proof line 9554). Its proof shows how to reduce to the case of a bounded complex of finite locally free modules of fixed rank, where the dual computation is available.
- The dual statement for a finite locally free module. Same chapter chow.tex, label lemma-chern-classes-dual (invoked at proof line 9557). It supplies the sign $(-1)^i$ for the Chern classes of the dual in the reduced case.
- The splitting principle for perfect objects, via Chern roots. Same chapter chow.tex, Remark \ref{remark-splitting-principle-perfect} (invoked at proof line 9559). It gives the alternative route: the Chern roots of $E^\vee$ are the negatives of those of E, which produces the same three signs.

Source reference. Original lemma 0FA5: sources/stacks-chow/pinned-original-source.tex, source lines 8520–8520. Navigation pointer for \ref{lemma-chern-classes-dual} used inside the counted span (line 9557); the argument that consumes it is retained in the span.

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Source reference. Original lemma 0FAC: sources/stacks-chow/pinned-original-source.tex, source lines 9394–9394. Navigation pointer for \ref{lemma-commutative-chern-perfect} used inside the counted span (line 9554); the argument that consumes it is retained in the span.

Source reference. Original remark 0FAD: sources/stacks-chow/pinned-original-source.tex, source lines 9494–9494. Navigation pointer for \ref{remark-splitting-principle-perfect} used inside the counted span (line 9559); the argument that consumes it is retained in the span.

Source reference. Original reference 02QL: sources/stacks-chow/pinned-original-source.tex, source lines 1372–1372. Navigation pointer for \ref{situation-setup} used inside the counted span (line 9546); the argument that consumes it is retained in the span.

Lemma 0.1. *In Situation 02QL let X be locally of finite type over S. Let $E \in D(\mathcal{O}_X)$ be a perfect object whose Chern classes are defined. Then $c_i(E^\vee) = (-1)^i c_i(E)$, $P_i(E^\vee) = (-1)^i P_i(E)$, and $ch_i(E^\vee) = (-1)^i ch_i(E)$ in $A^i(X)$.*

Proof. First proof: argue as in the proof of Lemma 0FAC to reduce to the case where E is represented by a bounded complex of finite locally free modules of fixed rank and apply Lemma 0FA5. Second proof: use the splitting principle discussed in Remark 0FAD and use that the chern roots of $E^\vee$ are the negatives of the chern roots of E. □